Kelvin Machaka

AI Engineer

Builds production AI, full-stack and geospatial systems: computer vision, predictive models, satellite data and the products around them.

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EXPERIENCE

AI Engineer · Earthasoft

2026 – present

Agritech platform for crop monitoring and agricultural insurance across Zimbabwe and southern Africa.

- Satellite crop monitoring: growth stages from optical imagery, radar to fill cloudy weeks, yield estimates and pest and disease risk.
- Insurer risk monitoring: claims, field evidence packs, damage maps, assessment and review steps, payouts and limits.
- Machine learning for field boundary detection and crop type detection.
- Full-stack delivery: Python services (FastAPI, Django, Celery), Next.js front end, PostgreSQL/PostGIS, Kubernetes releases, privacy-first product analytics.

SELECTED PROJECTS

Human-Centred Dispatch Orchestrator
Assigns field technicians to jobs by skill, distance and fatigue, predicts no-shows and job length, and keeps technicians working offline.
`Django REST · FastAPI · scikit-learn · Celery · Redis`

Tobacco Leaf Diagnosis
Photograph a tobacco leaf and get the likely disease and what to do about it, from a trained image model.
`TensorFlow / Keras · MobileNetV2 · FastAPI · Django REST · PostgreSQL`

Hospital Relief Platform
Lets donors pay verified hospital bills directly and see exactly how every dollar was used.
`Django REST · PostgreSQL · React · Tailwind · WeasyPrint`

FuelGuard
Catches fuel theft and fraud at filling stations by checking every pump sale against tank readings, in real time.
`Django REST · Celery · Django Channels · WebSockets · Redis`

Guard Scheduling and Patrols
Rosters security guards fairly and proves they were on site, with QR check-ins, GPS and live patrol maps.
`Django REST · Django Channels · Redis · PostgreSQL · React`

Cattle Disease Diagnosis
Combines a photo and the farmer's symptoms to suggest the likely disease, the treatment and a dose by body weight.
`PyTorch · ResNet-18 · Grad-CAM · FastAPI · Django REST`

SKILLS

AI / ML	Python, TensorFlow / Keras, PyTorch, scikit-learn, Model serving (FastAPI), Satellite imagery
Back end	Django REST, FastAPI, Node.js / Express, Celery, WebSockets (Channels), OpenAPI
Front end	TypeScript, React, Next.js, Tailwind CSS, Mapbox
Data & infra	PostgreSQL / PostGIS, MongoDB, Redis, Docker, Kubernetes, GitHub Actions

EDUCATION

Degree in Informatics and Analytics

AVAILABILITY

Available for freelance projects and remote roles.